SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO2: *Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)*

Assessment Tool Weightage: 50%

Duration: 1 Hour
QUESTIONS: 5 **Total Marks:50**

Annexure A

Scenario Based Assignment

Code of Conduct:

I Jabaraj T (192511187) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Scenario Based Assignment

1. Scenario : A government deploys AI-powered drones to deliver emergency medicines in a flood-affected region. The environment is highly dynamic—roads may suddenly become inaccessible, and weather conditions affect travel cost. The drone has access to: A heuristic estimate (straight-line distance), Partial real-time updates about blocked paths and Variable movement costs depending on terrain and weather. However, due to urgency, the drone must quickly decide routes with minimum delay and risk.

Tasks:

i. **Explain how Greedy Best-First Search would behave in this scenario and discuss its strengths and weaknesses under uncertainty**

ii. **Explain how A* would handle the same situation, including how it balances cost and heuristic**

iii. **Analyze the impact of heuristic accuracy on both algorithms**

iv. **Discuss how dynamic changes (blocked paths) affect search performance**

v. **Recommend the most suitable algorithm and justify your decision considering real-time constraints, optimality, and safety**

Answer:

i. Greedy Best-First Search
Greedy Best-First Search selects the path with the lowest heuristic value (estimated distance to the destination). It quickly finds a route, making it suitable for emergency situations. However, it ignores the actual travel cost, so it may choose blocked or risky paths and may not produce the optimal solution.

ii. A* Search
A* uses $f(n)=g(n)+h(n)$, where $g(n)$ is the actual cost travelled and $h(n)$ is the heuristic estimate. It balances travel cost and estimated distance, producing safer and more optimal routes while adapting better to varying terrain costs.

iii. Impact of Heuristic Accuracy
A more accurate heuristic improves both algorithms. Greedy becomes faster but still may not be optimal. A* becomes faster while maintaining optimality when the heuristic is admissible and consistent.

iv. Dynamic Changes
When roads become blocked, both algorithms must re-plan. Greedy may repeatedly choose poor alternatives, whereas A* recalculates using updated costs and usually finds a better alternative path.

v. Recommendation
A* is the most suitable because it balances speed, optimality, and safety. It efficiently handles changing costs and provides reliable routes for emergency medicine delivery.

2. Scenario: A smart city implements an AI system to optimize traffic signal timings across hundreds of intersections. The objective is to minimize Total waiting time, Fuel consumption and Traffic congestion. The system must continuously adjust signals based on traffic flow. However the search space is extremely large, traffic patterns change dynamically and the system may get stuck in locally optimal solutions

Tasks. Formulate this as an optimization problem and explain why traditional search is inefficient

- i. **Explain how Hill Climbing would work in this scenario and identify its limitations**
- ii. **Discuss issues like local maxima, plateaus, and ridges with real-world interpretation**
- iii. **Explain how Simulated Annealing or Genetic Algorithms can overcome these limitations**
- iv. **Recommend the best approach and justify your choice based on scalability and adaptability**

Answer:

Problem Formulation
Variables represent traffic signal timings. Objective: minimize waiting time, fuel consumption, and congestion while satisfying traffic rules. Traditional search is inefficient because the search space is extremely large.

i. Hill Climbing
Hill Climbing starts with a signal configuration and continuously moves to a neighboring solution that improves traffic flow. It is simple and fast but often gets trapped in local optima.

ii. Local Maxima, Plateaus and Ridges
Local maxima represent traffic settings that seem good but are not globally optimal. Plateaus occur when many configurations give similar performance. Ridges require multiple coordinated changes before improvement is seen.

iii. Simulated Annealing / Genetic Algorithm
Simulated Annealing occasionally accepts worse solutions to escape local maxima. Genetic Algorithms evolve multiple solutions using selection, crossover, and mutation, making them effective for large dynamic problems.

iv. Recommendation
Genetic Algorithms are the best choice because they scale well, adapt to changing traffic conditions, and search globally for high-quality solutions.

3. Scenario: An autonomous rover is deployed on Mars to explore unknown terrain. It must: navigate safely to collect samples, avoid hazards (craters, rocks), operate with limited sensing capability and make decisions without a complete map. The rover updates its knowledge as it explores, making this a real-time decision-making problem.

Tasks:

i. **Explain why this problem requires an online search agent instead of offline search**
ii. **Analyze the challenges of partial observability and dynamic environments**
iii. **Describe how the rover builds and updates its internal model**
iv. **Compare online search with uninformed and informed search strategies**
v. **Justify the most suitable approach considering uncertainty, adaptability, and safety**

Answer:

i. Online Search
The rover cannot know the complete environment beforehand, so it must make decisions while exploring using online search.

ii. Challenges
Limited sensing, unknown obstacles, and changing terrain make the environment partially observable and uncertain.

iii. Internal Model
The rover continuously updates its map using sensor data, storing safe paths, hazards, and explored locations.

iv. Comparison
Uninformed search explores blindly, informed search uses heuristics on known maps, while online search learns and plans simultaneously in unknown environments.

v. Recommendation
Online search is the most suitable because it adapts to uncertainty, improves safety, and supports real-time decision making.

4. Scenario: A university is designing an AI system to automatically generate exam timetables. The system must satisfy multiple constraints: No student should have overlapping exams, Limited number of exam halls, Certain subjects must be scheduled before others, Faculty availability constraints, Special accommodations for some students, The complexity increases as the number of courses and students grows.

Tasks :

Formulate the problem as a Constraint Satisfaction Problem (CSP)

i. **Clearly define variables, domains, and constraints with explanation**
ii. **Explain how backtracking search works in this scenario**
iii. **Discuss how constraint propagation (e.g., forward checking) improves efficiency**
iv. **Analyze real-world challenges and justify the best strategy for scalability**

Answer:

Problem Formulation
Variables: exam slots for each subject. Domains: available dates, times, and halls. Constraints: no student overlap, limited halls, subject order, faculty availability, and special accommodations.

i. Variables, Domains and Constraints
Each course is a variable. Domains are feasible slots. Constraints ensure all scheduling rules are satisfied.

ii. Backtracking Search
The algorithm assigns one exam at a time. If a constraint is violated, it backtracks and tries another assignment.

iii. Forward Checking
Forward checking removes invalid future choices after every assignment, reducing unnecessary search.

iv. Recommendation
Backtracking with forward checking and constraint propagation is the best approach because it efficiently handles large scheduling problems.

5. Scenario: Strategic Game AI for Real-Time Decision Making (Minimax & Alpha-Beta Pruning) A gaming company is developing an AI opponent for a real-time strategy game. The AI must complete against human players, Make decisions within strict time limits, Evaluate complex game states with multiple possible moves, The decision tree is extremely large, making computation expensive.

Tasks:

i. **Explain how the Minimax algorithm models this problem**
ii. **Analyze the computational challenges of large game trees**
iii. **Explain how Alpha-Beta pruning improves efficiency with detailed reasoning**
iv. **Design an evaluation function and justify the factors considered**
v. **Recommend a strategy for balancing optimality and real-time performance**

Answer:

i. Minimax
Minimax assumes both players play optimally. The AI chooses moves that maximize its score while minimizing the opponent's advantage.

ii. Computational Challenges
Large branching factors and deep game trees make Minimax computationally expensive for real-time games.

iii. Alpha-Beta Pruning
Alpha-Beta pruning eliminates branches that cannot influence the final decision, significantly reducing computation while producing the same optimal result as Minimax.

iv. Evaluation Function
The evaluation function considers unit strength, available resources, map control, health, strategic position, and possible future attacks.

v. Recommendation
Use Minimax with Alpha-Beta pruning and a strong heuristic evaluation function. This provides a good balance between optimal decisions and real-time performance.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand